

1 · Datasets

PlantVillage
18,160 tomato leaf images · 10 classes
single leaf, plain laboratory background

Tomato-Village
14,259 field images · 6 box-annotated classes
open fields, Rajasthan, India

2 · Data preparation

De-duplicate
hash every image, group
augmented copies

Leakage-free split
70 / 10 / 20, every copy
kept on one side

Resize
224² classification
800² detection

Leaf-safe augmentation
flips, mild colour jitter,
no lesion-destroying warps

3 · Shared trunk

Shared TomaBlock trunk
the same layer list in both models — trained on PlantVillage first, then reused to warm-start TomaNet-D

4 · Task heads

TomaNet-C — classification head
global average pool → FC + softmax
one disease name with a confidence score

TomaNet-D — detection head
FPN neck → detect heads at P3 / P4 / P5
many boxes, each with a class and a score

5 · Evaluation

Evaluation
accuracy · macro-F1 · Cohen’s κ · MCC · per class

Evaluation
precision · recall · mAP@0.5 · mAP@0.5:0.95 · per class